

```
1 import java.util.*;
2 class MatrixProduct
3 {
4     int arr[][] , r, c;
5     MatrixProduct(int r, int c)
6     {
7         this.r=r;
8         this.c=c;
9         arr=new int[r][c];
10    }
11
12    void accept()
13    {
14        Scanner sc = new Scanner(System.in);
15        System.out.println("Enter the Matrix");
16        for(int i=0; i<r; i++)
17        {
18            for(int j=0; j<c; j++)
19                arr[i][j]= sc.nextInt();
20        }
21    }
22
23    void display()
24    {
25        for(int i=0; i<r; i++)
26        {
27            for(int j=0; j<c; j++)
28                System.out.printf("%5d", arr[i][j]);
29            System.out.println();
30        }
31    }
32
33    MatrixProduct multiply(MatrixProduct A, MatrixProduct B)
34    {
35        MatrixProduct ob = new MatrixProduct(A.r, B.c);
36        for(int i=0; i<A.r; i++)
37        {
38            for(int j=0; j<B.c; j++)
39            {
40                for(int k=0; k<A.c; k++)
41                {
42                    ob.arr[i][j] += A.arr[i][k] * B.arr[k][j];
43                }
44            }
45        }
46        return ob;
47    }
48
49    void isValid(int c1, int r2)
50    {
```

```
51         if (c1 != r2)
52         {
53             System.out.println("Matrix Multiplication not possible ");
54             System.exit(0);
55         }
56     }
57     public static void main()
58     {
59         Scanner sc = new Scanner(System.in);
60         System.out.println("Enter the row and column for the 1st matrix");
61         int r1=sc.nextInt();
62         int c1=sc.nextInt();
63         MatrixProduct ob1 = new MatrixProduct(r1,c1);
64         ob1.accept();
65         System.out.println("First Matrix");
66         ob1.display();
67         System.out.println("Enter the row and column for the 2nd matrix");
68         int r2=sc.nextInt();
69         int c2=sc.nextInt();
70         ob1.isValid(c1,r2);
71         MatrixProduct ob2 = new MatrixProduct(r2,c2);
72         ob2.accept();
73         System.out.println("Second Matrix");
74         ob2.display();
75         MatrixProduct ob3= new MatrixProduct(r1,c2);
76         ob3=ob3.multiply(ob1,ob2);
77         System.out.println("Resultant Matrix");
78         ob3.display();
79     }
80
81 }
```